KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY (KIIT)

Deemed to be University U/S 3 of UGC Act, 1956

# DOMINO'S SALES PERFORMANCE ANALYSIS USING POWER BI

*A Business Intelligence Project Report*

Submitted by:

**Arsh Mohan Nishant**

**23052872**

B.Tech 3rd Year | Computer Science & Engineering

KIIT University, Bhubaneswar

Academic Year: 2025–26

# Abstract

This project presents a comprehensive analysis of Domino's pizza sales data using Microsoft Power BI. The objective is to understand sales performance, customer preferences, and ordering behavior through interactive data visualization techniques.

The dashboard is divided into two sections: an Overview Dashboard and a Detailed Analysis Dashboard. Key performance indicators such as total revenue, total orders, and total quantity sold are analyzed along with product performance, category contribution, and sales trends. Advanced visualization techniques including heatmaps are used to study customer behavior patterns.

This project demonstrates how business intelligence tools can transform raw data into actionable insights for better decision-making in the food and retail industry.

# Introduction

In today's competitive business environment, data-driven decision-making has become essential for organizational success. Businesses generate massive volumes of transactional data daily, and the ability to analyze and interpret this data effectively provides a significant competitive advantage.

Microsoft Power BI is a powerful business intelligence tool that enables users to connect to various data sources, transform raw data, and create interactive dashboards and visual reports. It allows stakeholders to explore data dynamically and derive meaningful insights without requiring deep technical expertise.

This project focuses on analyzing Domino's pizza sales data using Power BI. The aim is to evaluate business performance, identify top-selling products, and understand customer ordering patterns across different time periods, categories, and pizza sizes. By using a variety of visualizations, the dashboard provides a clear and structured view of the sales data that can assist business managers in making informed decisions.

# Problem Statement

Despite generating large volumes of daily transactional data, food businesses like Domino's often struggle to extract meaningful insights from their raw sales records. Key business questions remain unanswered:

- Which pizza categories and sizes contribute most to overall revenue?
- Which individual pizzas are the top performers in terms of sales?
- What are the peak ordering hours and days that drive the most business?
- How does revenue fluctuate across different months of the year?
- What is the average value of each customer order?

Without a structured analytical system, this data sits unused and business decisions are often made based on intuition rather than evidence. This project addresses that gap by building an interactive Power BI dashboard that transforms raw pizza sales data into clear, actionable visual insights — enabling data-driven decision-making for business growth and operational efficiency.

# `Objectives

The objectives of this project are:

- To analyze overall sales performance using key business metrics
- To calculate and visualize total revenue, total orders, and total quantity sold
- To identify top-performing pizzas based on revenue generation
- To analyze revenue distribution across different pizza categories
- To study the impact of pizza sizes on total revenue
- To observe monthly sales trends and identify seasonal fluctuations
- To understand customer ordering behavior based on time of day and day of week
- To compute average order value as a measure of customer spending
- To create an interactive and user-friendly dashboard for business stakeholders

# Dataset Description

The dataset used in this project contains detailed transactional information about pizza sales from Domino's. It includes the following key attributes:

- pizza_id — Unique identifier for each pizza item in an order
- order_id — Unique identifier for each customer order
- pizza_name_id — Short code representing the pizza name
- quantity — Number of pizzas ordered
- order_date — Date on which the order was placed
- order_time — Time at which the order was placed
- unit_price — Price of a single pizza
- total_price — Total price for the line item
- pizza_size — Size of the pizza (S, M, L, XL, XXL)
- pizza_category — Category of the pizza (Chicken, Classic, Supreme, Veggie)
- pizza_name — Full name of the pizza
- Hour, Day, Day Number — Derived columns for time-based analysis

The dataset consists of approximately 48,620 records spanning the year 2015. This large dataset enables meaningful statistical analysis of sales patterns and customer behavior. The data was imported into Power BI Desktop in CSV format and used for creating calculated measures and visualizations.

***Table 1: Sample Dataset Records***

| pizza_id | order_id | pizza_name | quantity | pizza_size | pizza_category | unit_price | total_price |
|---|---|---|---|---|---|---|---|
| 6 | 2 | The Thai Chicken Pizza | 1 | L | Chicken | 20.75 | 20.75 |
| 42 | 16 | The Thai Chicken Pizza | 1 | L | Chicken | 20.75 | 20.75 |
| 68 | 25 | The Thai Chicken Pizza | 1 | L | Chicken | 20.75 | 20.75 |
| 164 | 71 | The Thai Chicken Pizza | 1 | L | Chicken | 20.75 | 20.75 |

# Tools and Technologies Used

## Microsoft Power BI Desktop

The primary tool used for this project. Power BI Desktop was used to import the dataset, perform data transformations, create DAX measures, and build the interactive dashboard with multiple visualizations and slicers.

## DAX (Data Analysis Expressions)

DAX was used to create calculated measures and derived columns. Key measures created include Total Revenue, Total Orders, Total Quantity Sold, and Average Order Value. DAX also enabled time-based calculations such as extracting hour and day from order timestamps.

## CSV (Comma-Separated Values)

The raw dataset was provided in CSV format. Power BI's built-in data connectors were used to import and load the CSV file directly into the data model.

# Methodology

The project was carried out in the following stages:

**Step 1 — Data Import**

The pizza sales CSV dataset was imported into Power BI Desktop using the 'Get Data' feature. The data was loaded into the Power Query Editor for initial inspection.

**Step 2 — Data Cleaning & Transformation**

Basic data cleaning was performed to ensure accuracy and consistency. This included verifying data types, removing null values, and formatting date and time columns correctly. Derived columns such as Hour, Day, and Day Number were extracted from the order timestamp for time-based analysis.

**Step 3 — DAX Measure Creation**

Key business metrics were calculated using DAX expressions. Measures such as Total Revenue (SUM of total_price), Total Orders (DISTINCTCOUNT of order_id), Total Quantity Sold (SUM of quantity), and Average Order Value were created and validated.

**Step 4 — Visualization Development**

Multiple visualization types were used to represent different aspects of the data. These include KPI cards, bar charts, pie charts, line charts, heatmap tables, and column charts. Each visual was configured with appropriate axes, legends, and data labels for clarity.

**Step 5 — Dashboard Design**

The dashboard was organized into two pages. Page 1 provides a high-level overview of sales performance, while Page 2 offers a detailed behavioral analysis. Interactive slicers were added to allow category-based filtering, enhancing the user experience.

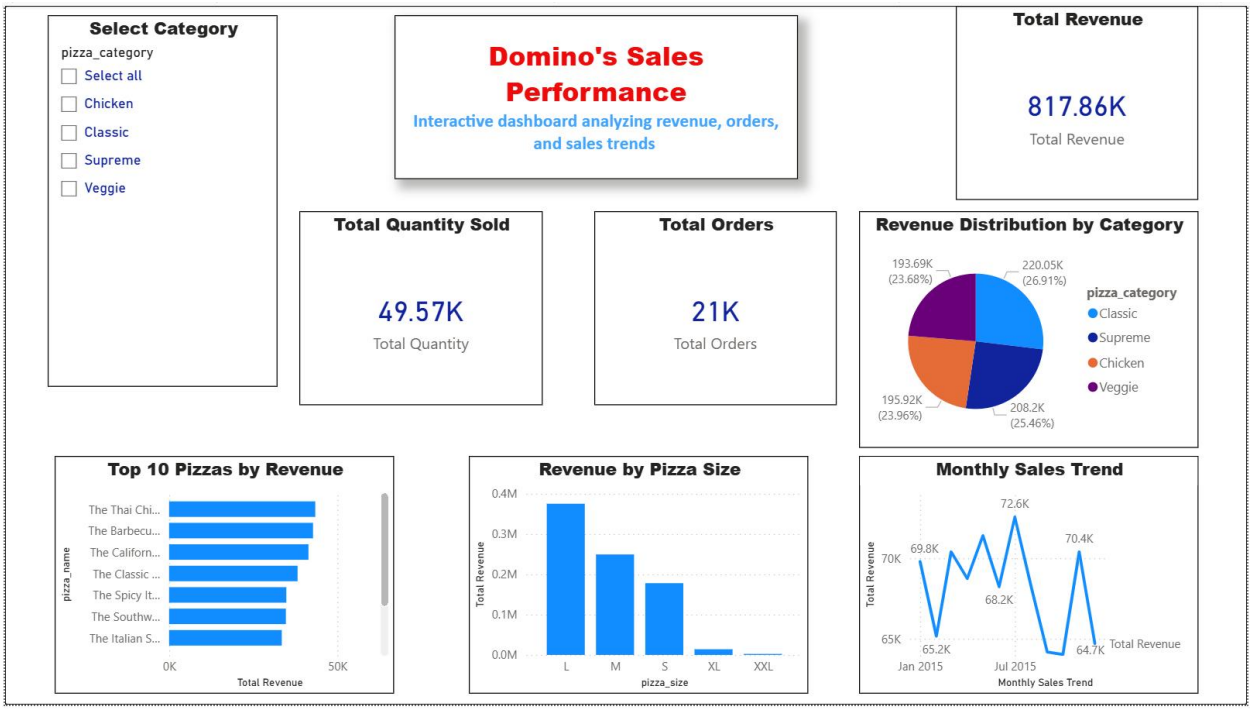
# Dashboard Analysis

## Page 1 — Sales Performance Overview

The first page of the dashboard provides a comprehensive overview of Domino's sales performance. It contains multiple visualizations organized for quick executive-level insights.

A category slicer on the left panel allows users to filter all visuals by pizza category: Chicken, Classic, Supreme, and Veggie. This interactive filter updates all charts simultaneously for focused analysis.

***Figure 1: Dashboard Page 1 — Sales Performance Overview***




The three KPI cards at the top display the most critical business metrics at a glance — Total Revenue of ₹817.86K, Total Orders of 21K, and Total Quantity Sold of 49.57K.

The Revenue Distribution by Category pie chart reveals that Supreme (26.91%) and Chicken (25.46%) categories are the top contributors, closely followed by Classic (23.96%) and Veggie (23.68%). This fairly even distribution suggests that all categories are performing well with no single category dominating the market.

The Top 10 Pizzas by Revenue bar chart identifies the best-selling individual pizzas, with The Thai Chicken Pizza leading in revenue generation. This information is critical for inventory planning and promotional strategies.

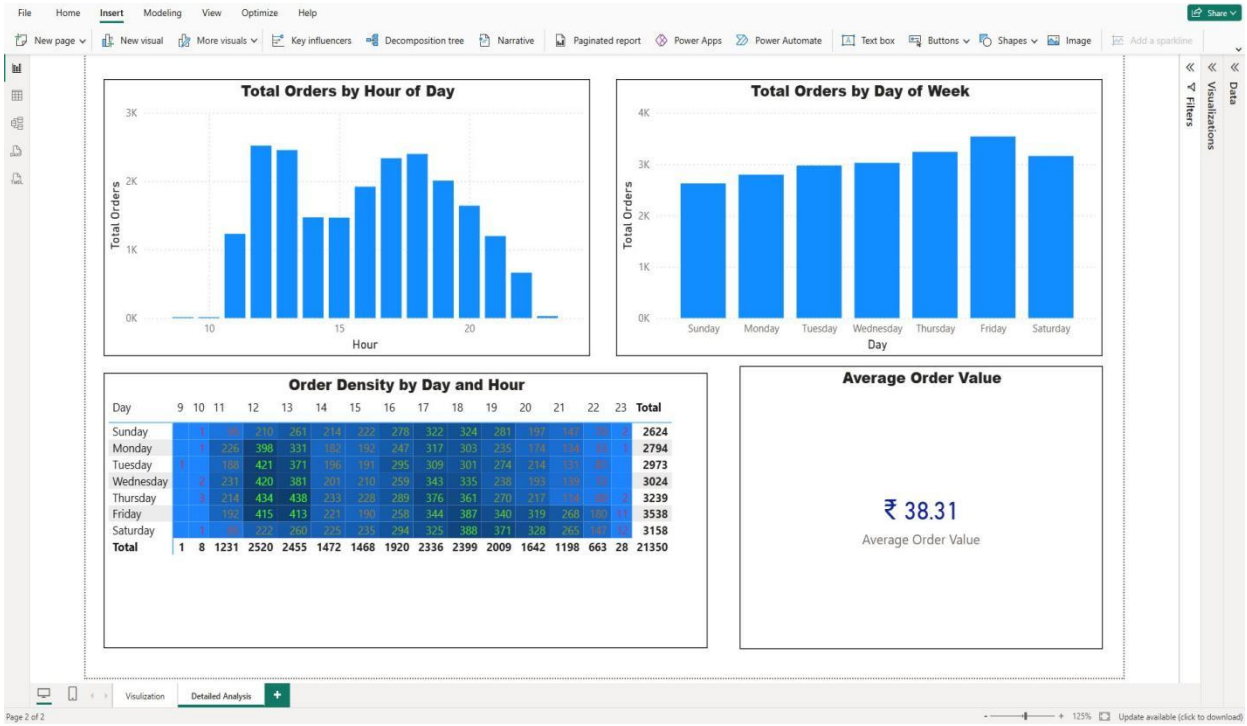
The Revenue by Pizza Size column chart shows that Large (L) pizzas generate the highest revenue, followed by Medium (M) and Small (S). XXL pizzas have minimal contribution due to limited orders.

The Monthly Sales Trend line chart shows revenue fluctuations throughout the year, with peaks observed in July (₹72.6K) and notable dips in certain months. This trend helps management plan promotions during slower periods.

## Page 2 — Detailed Behavioral Analysis

The second page provides a deeper analysis of customer ordering behavior, focusing on temporal patterns that can inform staffing, marketing, and operational decisions.

***Figure 2: Dashboard Page 2 — Detailed Behavioral Analysis***



The Total Orders by Hour of Day bar chart reveals that peak ordering times occur around noon (12–13) and early evening (17–18), corresponding to lunch and dinner hours. This insight can be used to optimize kitchen staffing during these periods.

The Total Orders by Day of Week bar chart shows that Friday (3,538 total orders) is the busiest day of the week, followed by Thursday and Saturday. Sundays tend to have the lowest order volume. This pattern aligns with typical consumer behavior where people prefer to order food on weekends and at the end of the work week.

The Order Density by Day and Hour heatmap table is the most insightful visualization in the dashboard. It displays a matrix of order counts for each combination of day and hour, with color intensity indicating order density. This allows management to identify the exact busiest time slots — for example, Fridays at 12:00 (415 orders) and 13:00 (413 orders) are peak periods.

The Average Order Value KPI card shows ₹38.31, indicating the typical spending per customer order. This metric is useful for tracking revenue efficiency and setting pricing strategies.

## Key Insights and Findings

The following key insights were derived from the Power BI dashboard analysis:

***Table 2: Key Performance Indicators Summary***

| KPI Metric | Value | Description |
|---|---|---|
| Total Revenue | **₹ 817,860** | Combined revenue from all pizza orders |
| Total Orders | **21,000** | Total number of pizza orders placed |
| Total Quantity Sold | **49,570** | Total number of pizzas sold across all categories |
| Average Order Value | **₹ 38.31** | Average revenue generated per order |

- The Thai Chicken Pizza is the top-revenue generating product, making it the most popular pizza by sales.
- Large-sized pizzas contribute the most to total revenue, suggesting customer preference for larger portions.
- All four pizza categories (Chicken, Classic, Supreme, Veggie) contribute roughly equally to total revenue, indicating a balanced product portfolio.

- Peak ordering hours are between 12:00–13:00 (lunch) and 17:00–18:00 (dinner), providing clear staffing guidance.
- Friday is consistently the busiest day of the week with 3,538 total orders, while Sunday records the least activity.
- Monthly revenue peaks in July at ₹72.6K, with some months showing dips that could benefit from targeted promotions.
- The average order value of ₹38.31 provides a benchmark for pricing and upselling strategies.
-

# Conclusion

This project successfully demonstrates the power of Microsoft Power BI as a business intelligence tool for analyzing and visualizing complex sales data. By importing and transforming a dataset of over 48,000 pizza sales records, meaningful insights were extracted through interactive dashboards that go far beyond what traditional spreadsheet analysis can offer.

The two-page dashboard provides both high-level executive summaries and granular behavioral insights. Business stakeholders can use these visualizations to make data-driven decisions regarding product strategy, marketing campaigns, staffing schedules, and pricing models.

The project also highlights the importance of time-based analysis in retail — understanding when customers order (peak hours and days) is just as important as knowing what they order. The Order Density heatmap, in particular, is a powerful tool for operational planning.

Overall, this project reinforces that business intelligence tools like Power BI can transform raw transactional data into a competitive advantage, enabling organizations to move from reactive to proactive decision-making.

# References


- Microsoft Power BI Documentation — https://docs.microsoft.com/en-us/power-bi/
- DAX Reference Guide — Microsoft Learn
- Pizza Sales Dataset — Kaggle (pizza_sales.csv)
- Power BI Desktop — Version used for dashboard development